

Exercices d'entraînement partie 1

Corrigé exercice 52 :

1. $\operatorname{Re}(z_1) = 2$ et $\operatorname{Im}(z_1) = 0$.
2. $\operatorname{Re}(z_2) = 0$ et $\operatorname{Im}(z_2) = -3$.
3. $\operatorname{Re}(z_3) = -3$ et $\operatorname{Im}(z_3) = 1$ car $z_3 = -3 + i$.
4. $z_4 = z_1 + z_3 = 2 - 3 + i = -1 + i$ donc $\operatorname{Re}(z_4) = -1$ et $\operatorname{Im}(z_4) = 1$.
5. $z_5 = z_2 \times z_3 = -3i(-3 + i) = 9i + 3 = 3 + 9i$ donc $\operatorname{Re}(z_5) = 3$ et $\operatorname{Im}(z_5) = 9$.

Corrigé exercice 54 :

1. $z_1 = 3 - 2i - 3 - 2i = -4i$
2. $z_2 = -2 + 2i - 2 - i = i$
3. $z_3 = 3 - 2 + i(2 + 3) = 1 + 5i$
4. $z_4 = 1^3 - 3i + 3i^2 - i^3 = 1 - 3i - 3 + i = -2 - 2i$
5. $z_5 = 1^5 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5 = 1 - 5i - 10 + 10i + 5 - i = -4 + 4$